# Beyond the First Visit

CMAT attendance and mathematics outcomes. Paper 2 - provisional scientific synthesis, 26 October 2026. For research faculty and CMAT administration.

Source: Consolidation plan

# Principal message

The strongest evidence distinguishes zero recorded visits from initial use. Adjusted outcome differences are sizeable, while positive-frequency increments are less precise and model-dependent. No causal benefits are identified.

Source: Paper 2.1 Wald/Holm and 10_benchmark

# Why mathematics support matters

Learning centres supplement instruction amid different levels of preparation. Use is self-selected; non-engagement may reflect access barriers. Experiments and observational studies provide different types of evidence.

Source: Lawson et al. 2020; Mullen et al. 2024; Pugatch & Wilson 2018

# CMAT and the optional PPA1 activity

PPA1 provides one optional three-CMAT-visit integration activity. The study does not observe individual intentions, PPA eligibility or credit use. Three visits motivate an exploratory grouping, not a discontinuity design.

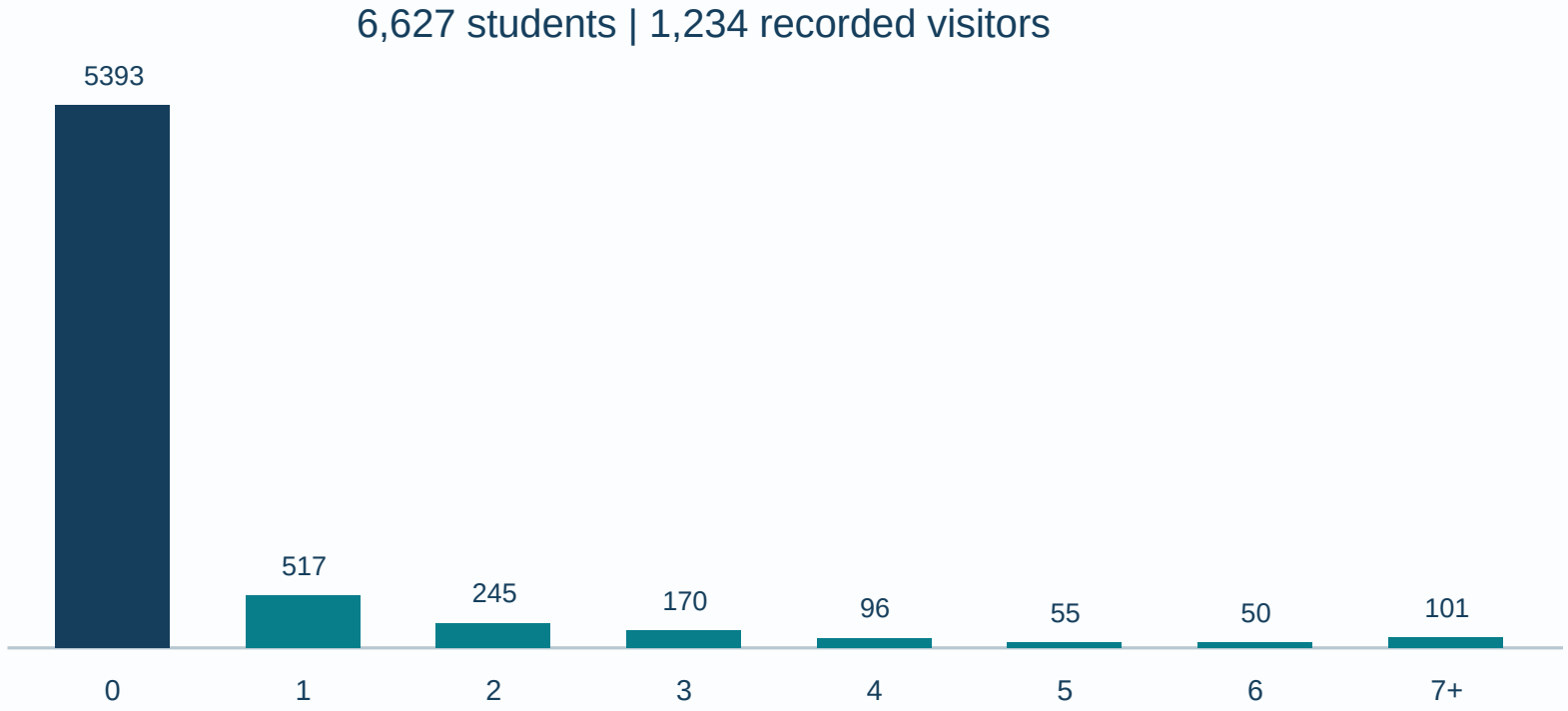
Source: Consolidation plan, 2026-10-26

# Cohort and empirical design

First eligible MU attempt with CMAT coverage, context fixed effects, degree adjustment and professor-period clustered uncertainty. Primary outcomes: Z_GRADE_PRIMARY, PASS; numeric-only and administrative sensitivities differ in sample and target.

Source: Paper 2.1 specifications; consolidation plan

# Visit distribution and precision

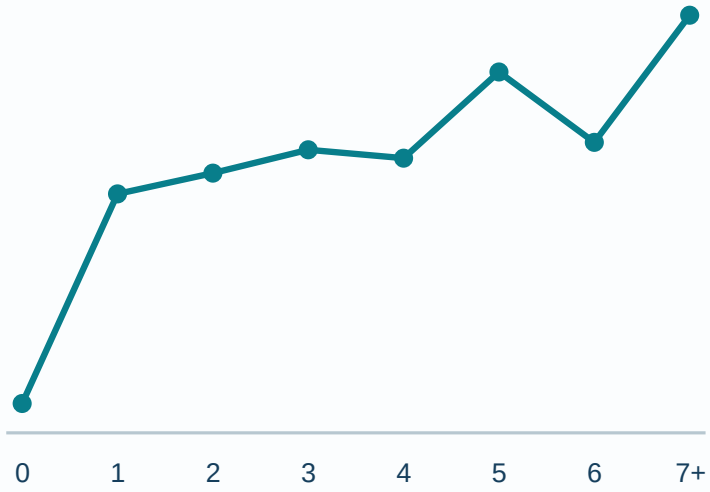



N=6,627; 190 professor-period contexts; 5,393 with zero visits and 1,234 with positive recorded visits. Sparse high-frequency support constrains fine-grained contrasts.

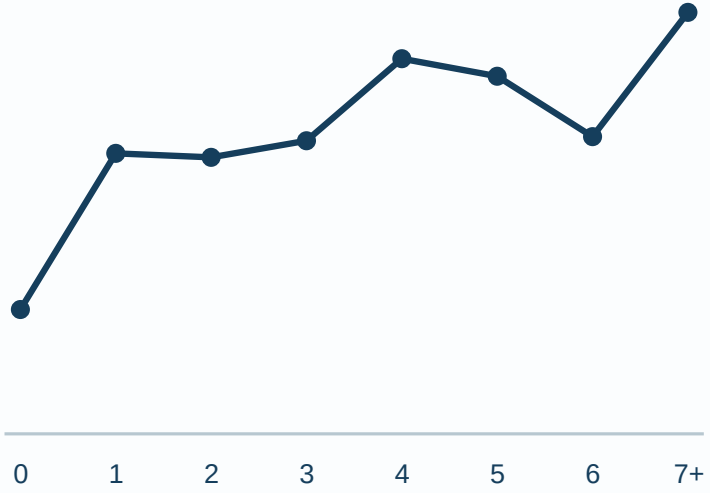
Source: 01_cohort_flow.csv; 10n_zero_inclusive_7plus_descriptives.csv

# Raw grades and passing rates

Raw standardized grade (SD)




Raw PASS rate (%)




These are unadjusted between-student means, not longitudinal growth or experimental effects. The exact-frequency series is not strictly monotone.

Source: 10n_zero_inclusive_7plus_descriptives.csv

# Any attendance vs zero

Adjusted 1+ minus 0: +0.361 grade SD, CI [0.302,0.421], and +15.29 PASS percentage points, CI [12.44,18.15]. The 1+ pool is not exactly one visit.

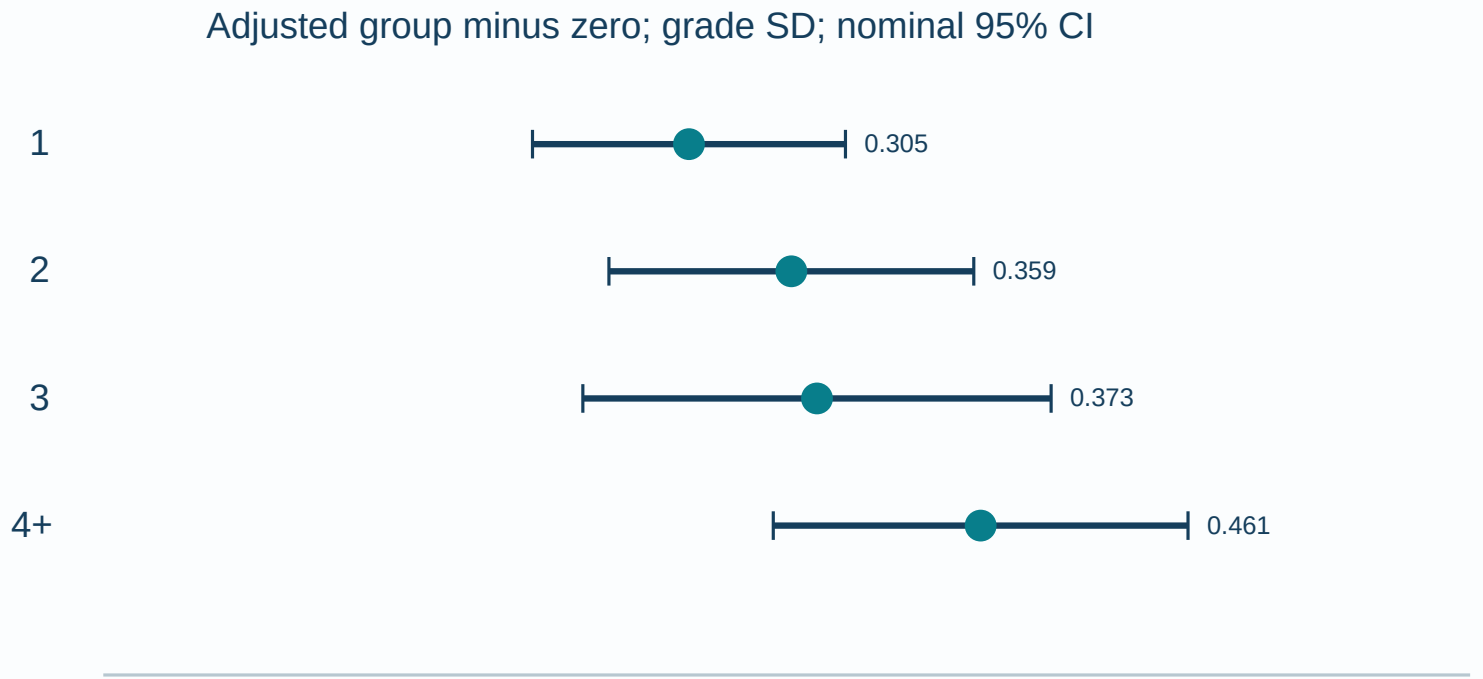
Source: 10_benchmark_0_vs_1plus.csv

# Wald and Holm inference

Original primary Paper 2.1 users-only analysis: 1/2/3/4/5/6+ with 15 Holm-adjusted pairs per outcome; none survives. The omnibus tests do not reject. Complementary full-cohort analysis: 0/1/2/3/4+ with 10 pairs per outcome; zero-versus-positive differences are clearer. Separate sensitivity families: users-only seven groups (21 pairs), zero-inclusive eight groups (28 pairs), and exploratory adjacent comparisons (7 pairs). SEs are clustered by professor-period; nominal 95% CIs are not Holm simultaneous CIs.

Source: Paper 2.1 tables 12-14, 21-23, 33/41, 26a/26k; methods PR #97; QA PR #96

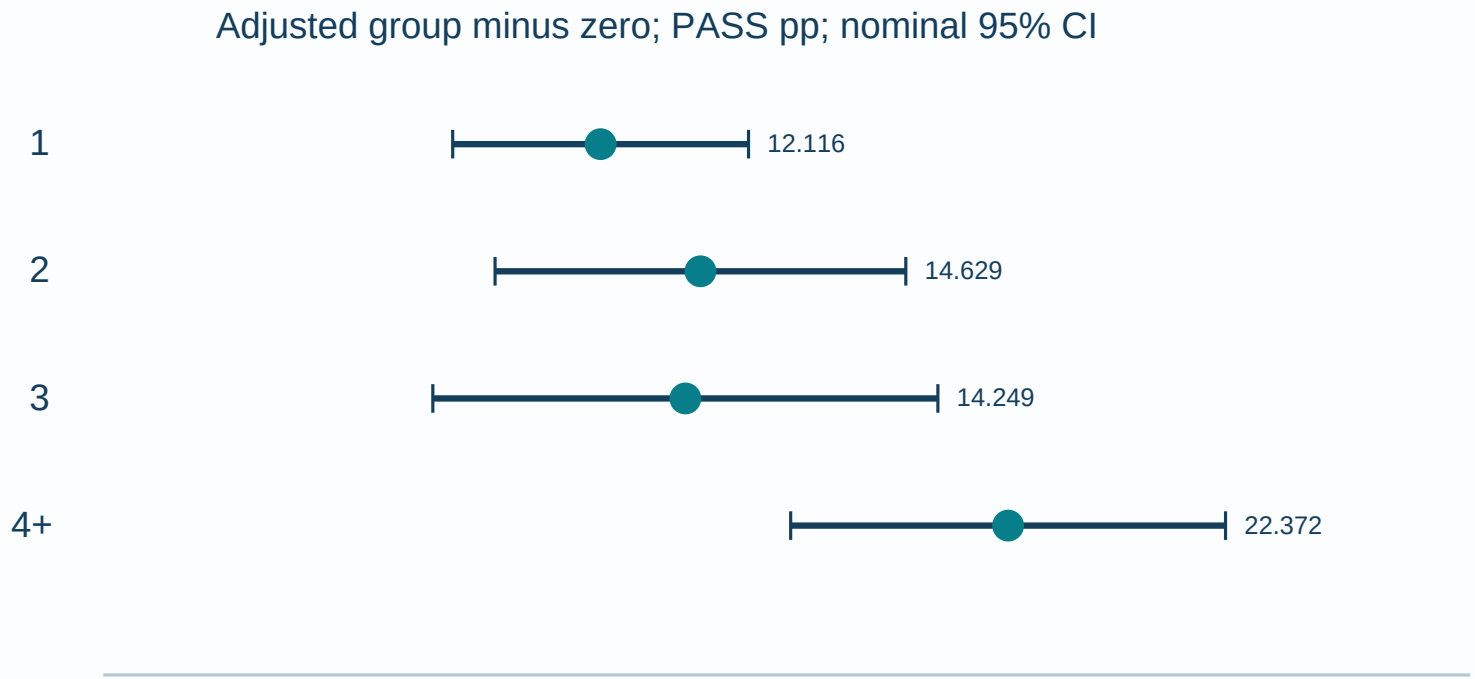
# Adjusted standardized-grade contrasts




This figure is the complementary ten-pair zero-inclusive model. Every group differs from zero after Holm; no positive-versus-positive grade contrast survives here or in the original users-only 15-pair family.

Source: 33_exact_visit_groups_fe_pairwise.csv, Paper 2.1

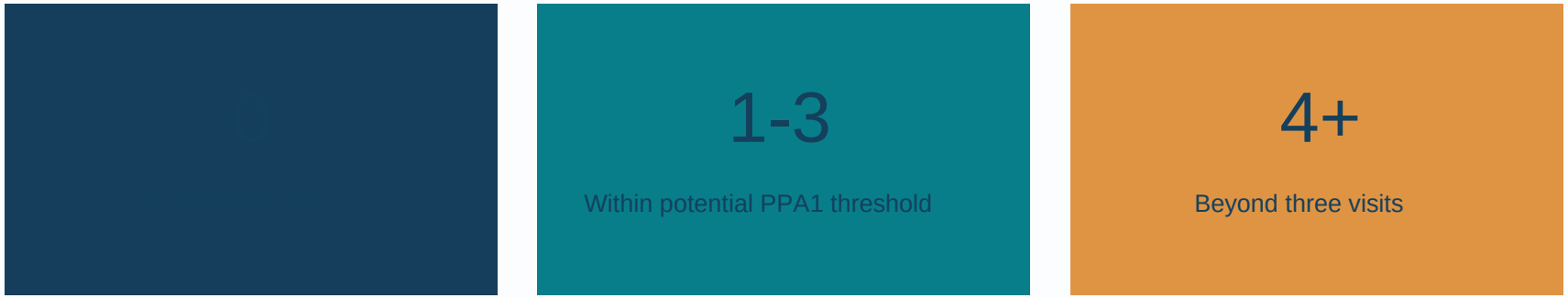
# Adjusted PASS contrasts




In the complementary ten-pair model, zero differs from each positive group. 1 versus 4+ also survives Holm (p~0.011). The original users-only 15-pair family has no surviving pairwise contrast.

Source: 41_exact_visit_groups_pass_fe_pairwise.csv, Paper 2.1

# PPA grouping: not yet validated




Conceptual categories - PPA analysis pending validation

Primary 0 | 1-3 | 4+; four-group sensitivity 0 | 1-3 | 4 | 5+. Pending validated adjusted estimates. The figure is a grouping diagram, not a plotted result.

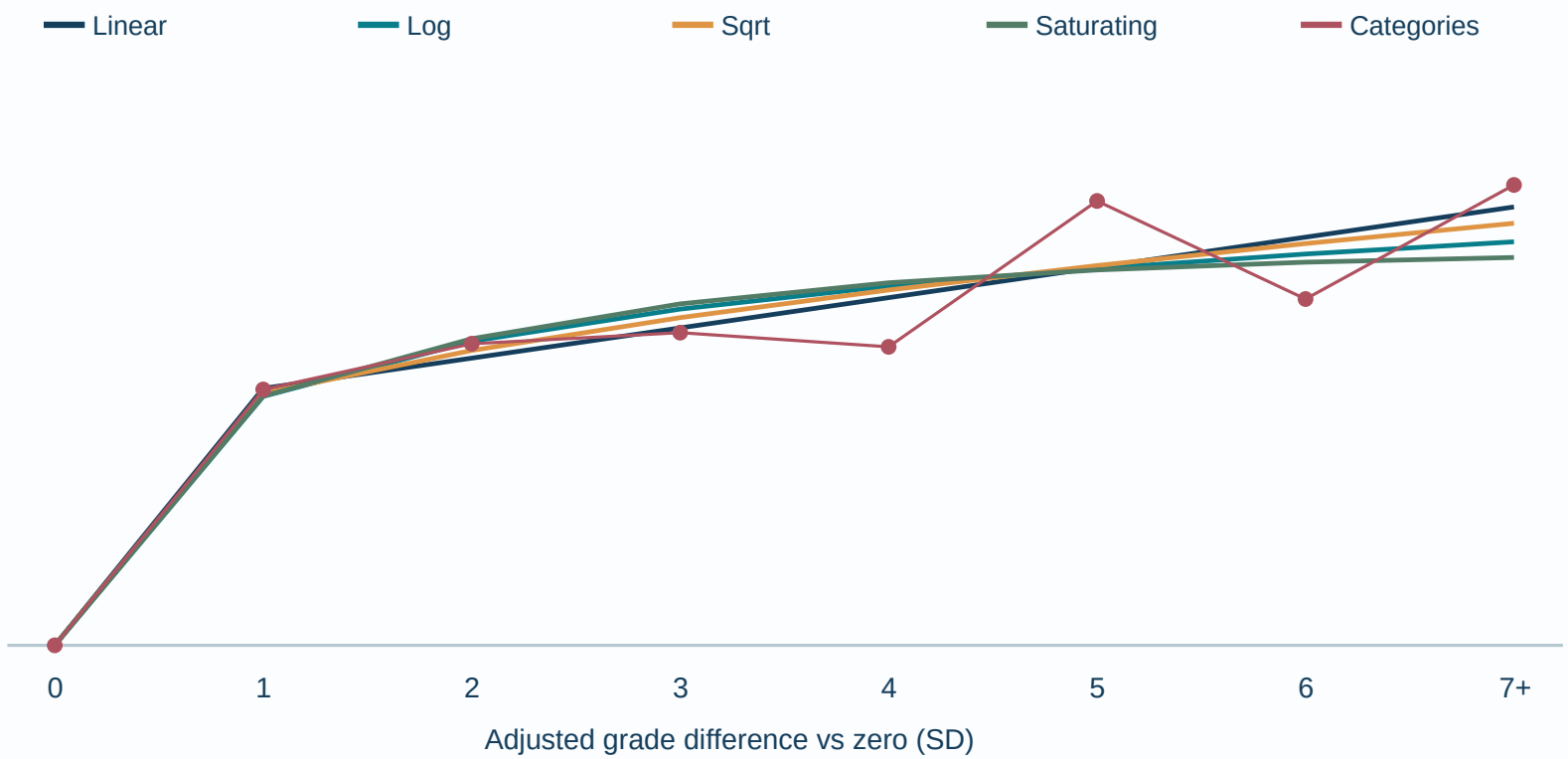
Source: PAPER2_CONSOLIDATION_PLAN.md

# Two-part first/additional visits

Adjusted working model: Z first-visit +0.307 SD, additional-visit slope +0.036 SD; PASS first +11.92 pp, extra +2.23 pp. The common post-first slope is imposed and top-coded at 7+.

Source: two_part_attendance_z_pass.csv

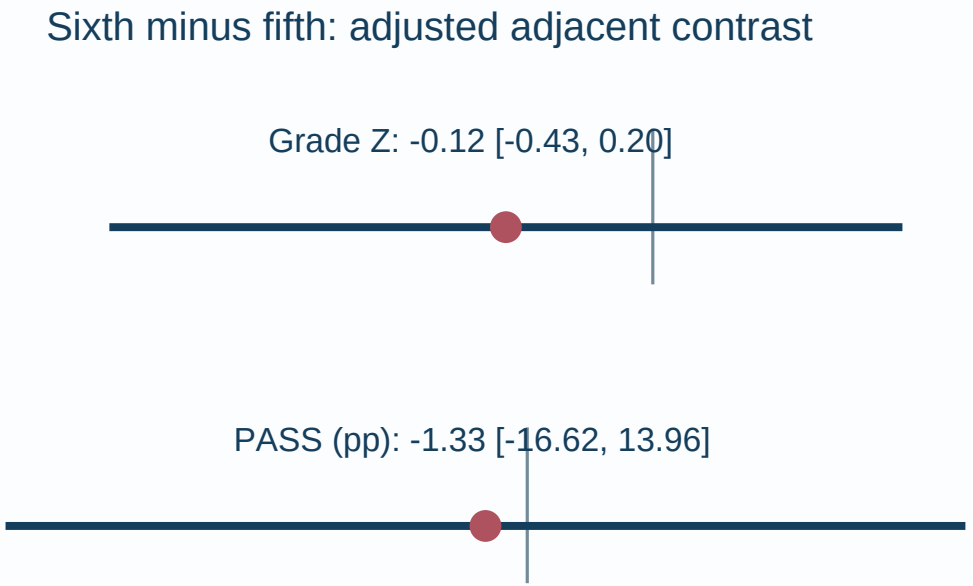
# Five fitted curve families




Linear, log, square-root and saturating fits impose different increment shapes; categorical effects do not. Slightly better in-sample AIC or grouped validation does not prove causal diminishing returns.

Source: attendance_curves/adjusted_curves_cluster_ci.csv

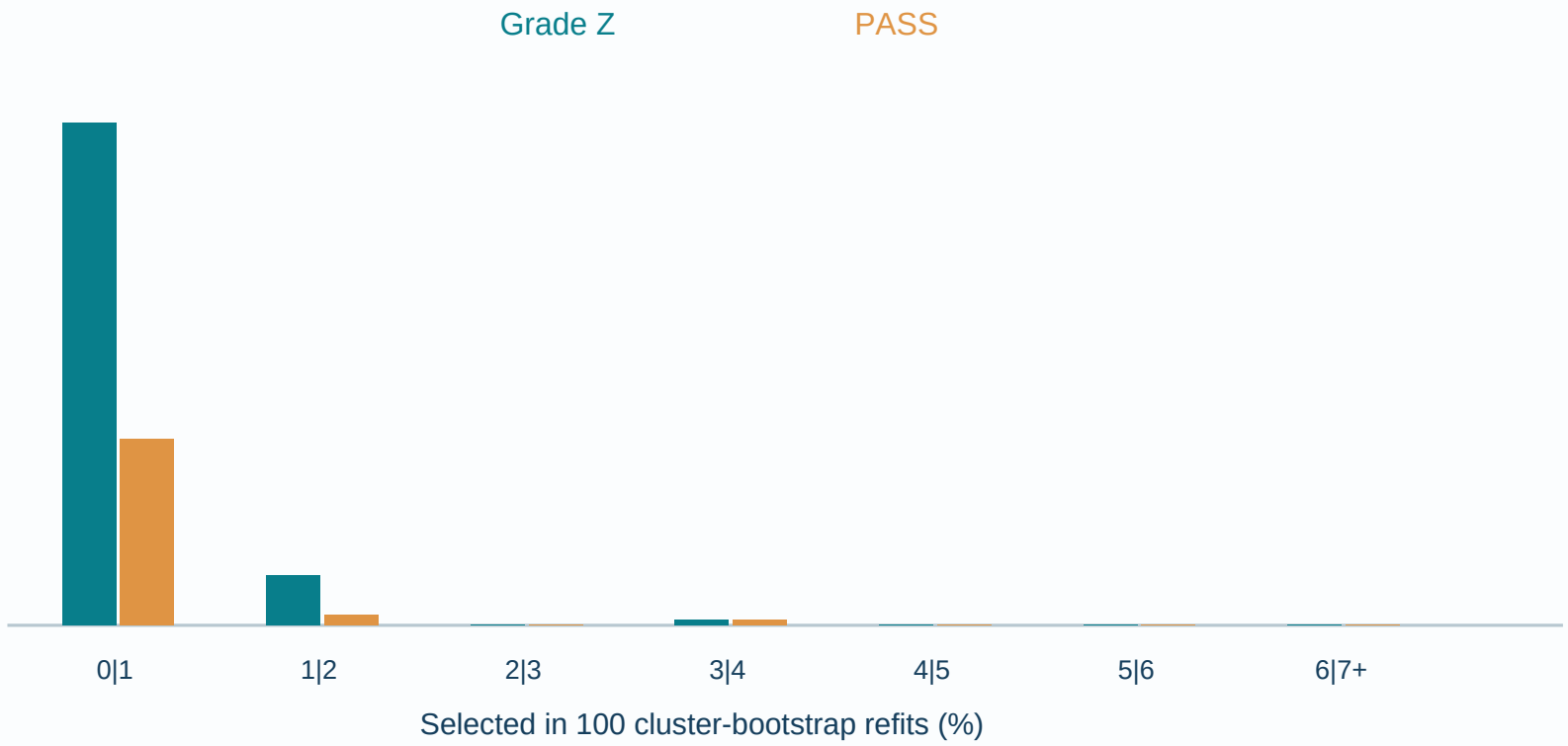
# Five-to-six dip: broad uncertainty




Sixth minus fifth adjusted contrast: grade -0.117 SD, CI [-0.433,+0.199]; PASS -1.33 pp, CI [-16.62,+13.96]. Neither survives seven-adjacent Holm. N5=55, N6=50.

Source: attendance_curves/adjacent_differences_holm.csv

# Fused-lasso boundary stability




Grade discovery selects [0] | [1-7+]; 0|1 boundary appears in 100/100 cluster bootstrap refits. PASS one-SE-style selection fuses all counts, although its 0|1 boundary appears 37/100 times.

Source: fused 7plus blocks and boundary_stability CSVs

# Internally held-out grade comparison

Reserved n=1,952 in 57 contexts: discovered Z group 1+ minus zero +0.264 SD, CI [0.150,0.377]. PASS has no selected boundary, so no comparable selected Wald contrast. Not an external or prospectively untouched replication.

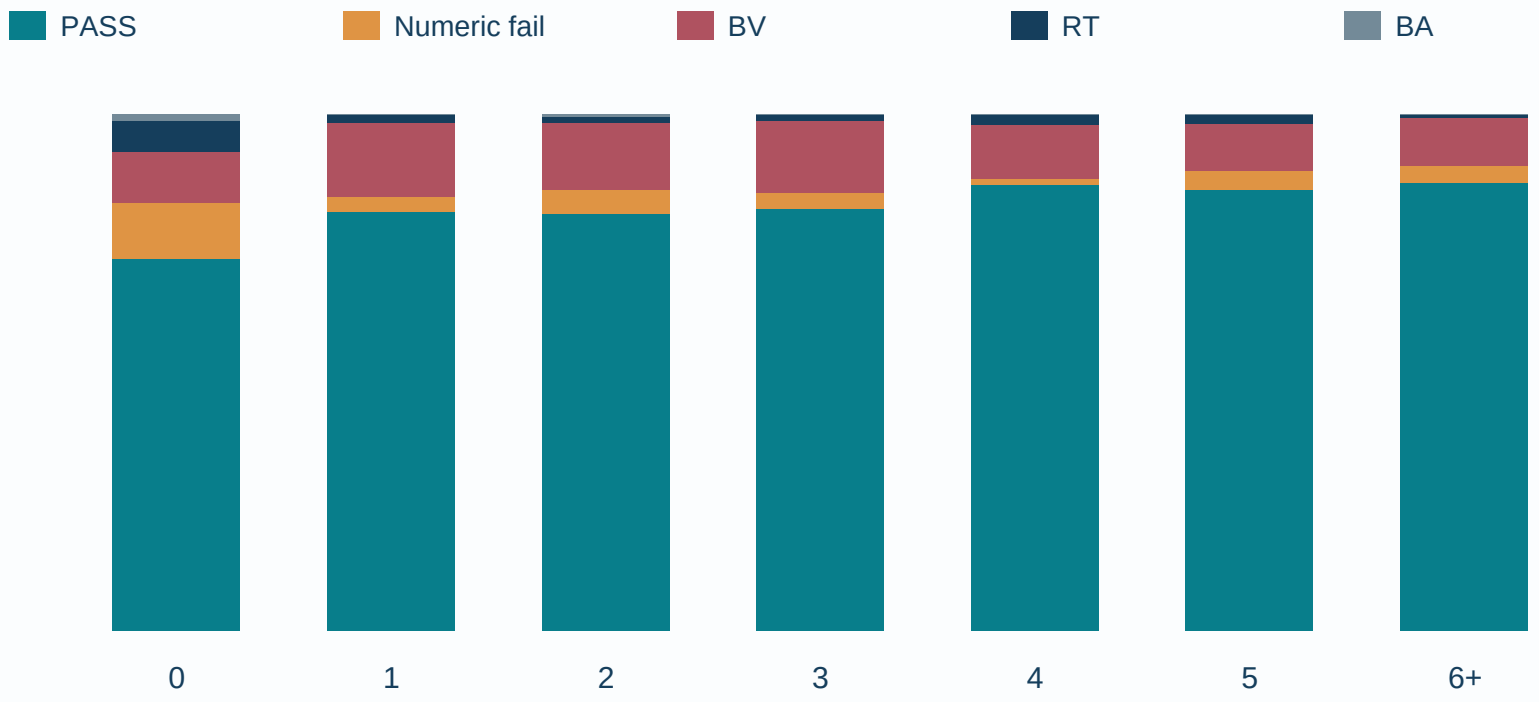
Source: 7plus_z_grade_primary_holdout_wald.csv

# Mixtures and skew-normal alternatives

Historical EM Gaussian mixtures and skew-normal fits test distributional adequacy under different outcome encodings. A better K=2 likelihood does not identify two student types or even two visible modes. Prospective modality test remains pending.

Source: Distributional branch results/paper21/tables/37-52

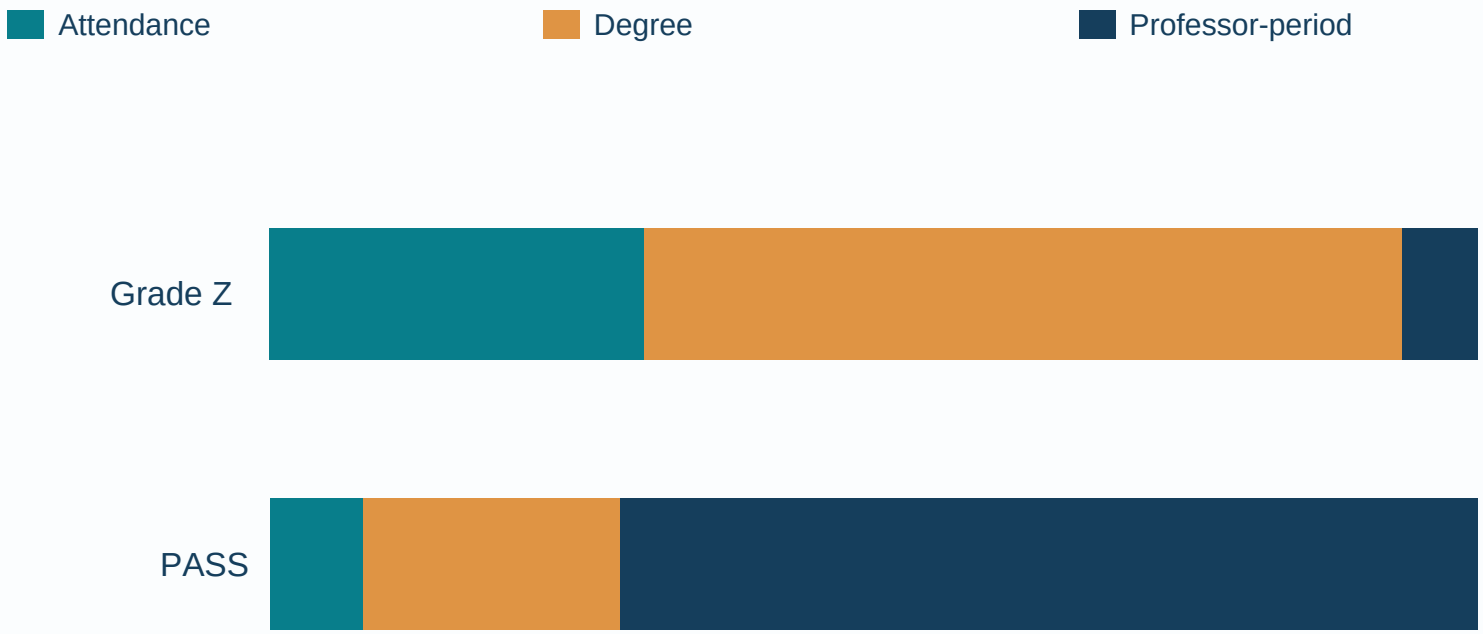
# Observed outcomes and administration




PASS, numeric fail, BV, RT and BA are different observed states. Numeric-only grades exclude administrative outcomes; primary Z applies defined imputation. Combining these populations changes the estimand.

Source: 10d_zero_inclusive_exact_administrative_composition.csv

# In-sample Shapley fit attribution

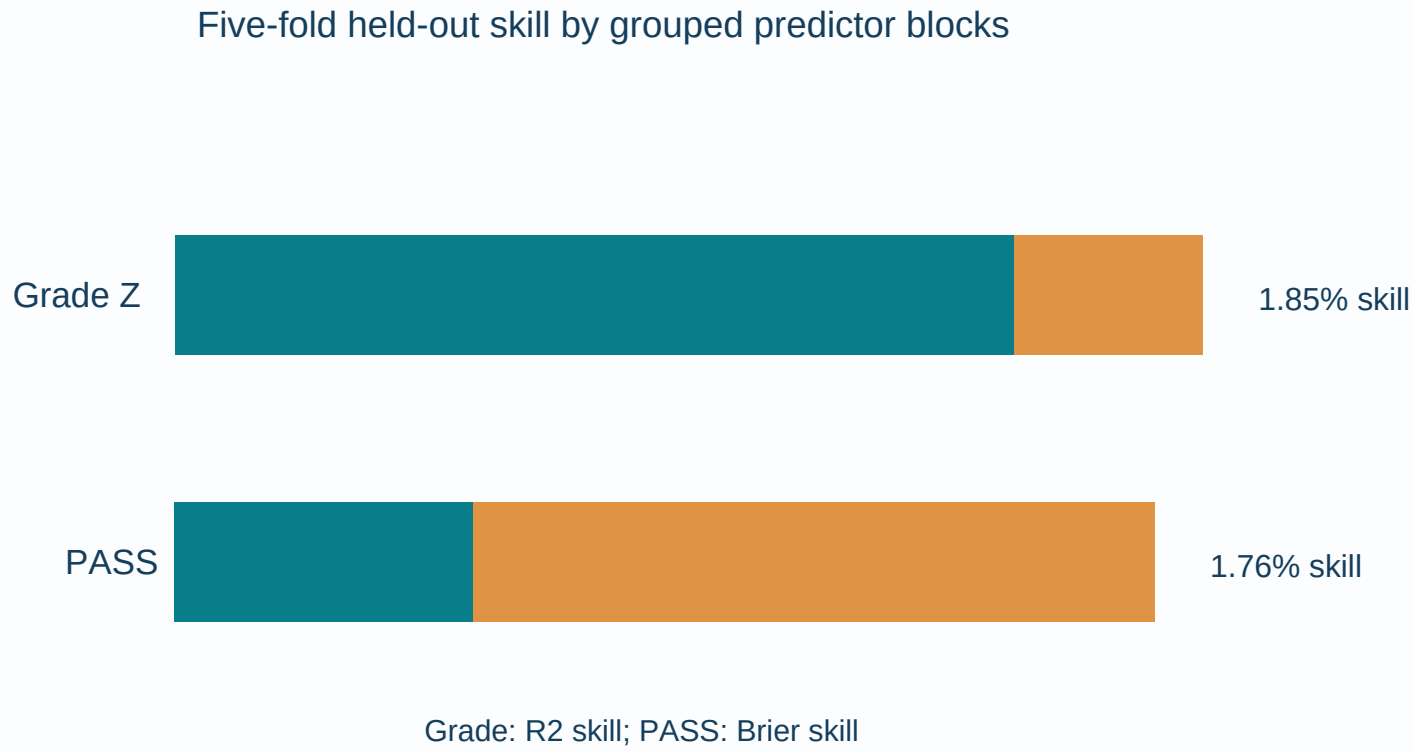



Share of in-sample fit, not causal importance

Model D: attendance accounts for approximately 31% of grade R2 and 8% of PASS R2. Degree dominates grade fit; professor-period dominates PASS fit. These are not causal contributions.

Source: shapley_importance/in_sample_shapley_12_models.csv

# Out-of-context predictive skill




Five-fold professor-period-held-out analysis: grade R2 skill 1.85%, PASS Brier skill 1.76%. Test-context FE indicators are unavailable. These are retrospective predictions, not actionable start-of-term warnings.

Source: shapley_importance/out_of_context_shapley.csv

# Synthesis and next study

Supported: large adjusted zero/positive separation, uncertain consecutive increments, model-dependent selected boundaries and differing administrative outcome composition. Unresolved: self-selection, PPA1 participation, preparation, functional form and prospective generalization. Next: validate primary and secondary PPA specifications; harmonize denominators; collect baseline readiness and visit timing; evaluate ethical prospective encouragement. Release gate - DO NOT RELEASE: independent QA PR #96 reports a critical public controlled-data exposure and invalid PASS-logit significance flags. Institutional response and renewed review are prerequisites.

Source: Methods PR #97, QA PR #96 and committed aggregates, 2026-10-26

# Appendix A - evidence hierarchy

Inferential anchors: original user-only 15-pair Wald/Holm contrasts and complementary zero-inclusive ten-pair contrasts. Sensitivity analyses: positive-only tail 21-pair and zero-inclusive 28-pair families; first/additional visit and fitted curve shapes. Model diagnostics: fused-lasso grouping; Gaussian mixtures versus skewness; Shapley in-sample attribution; retrospective held-out prediction. PPA: mutually exclusive 0 | 1-3 | 4+ and 0 | 1-3 | 4 | 5+ remain exploratory and not estimated on authorized institutional microdata.

Source: Methods claims-to-evidence matrix (PR #97); PPA PR #98

# Appendix B - cohort numbers

Eligible first MU attempts across all historical periods: 7,777; covered analysis: 6,627. Exact n: 0=5393, 1=517, 2=245, 3=170, 4=96, 5=55, 6=50, 7+=101. 190 professor-period contexts.

Source: 01_cohort_flow.csv; 10n_zero_inclusive_7plus_descriptives.csv

# Appendix C - multiplicity

For a pair i,j, test H0: adjusted mu(i) = mu(j) with professor-period cluster-robust Wald uncertainty. Holm step-down adjustment is applied within a declared comparison family. Paper 2.1 families, per outcome: primary users-only 15; users-only 7-level sensitivity 21; zero-inclusive five-group 10; zero-inclusive eight-group 28. Other analyses: seven adjacent steps, two piecewise terms; proposed PPA-18 = (3+6) x 2 outcomes in one family.

Nominal cluster-robust intervals are not simultaneous Holm intervals.

Source: Paper 2.1 tables 12-14, 21-23, 33/41, 26a/26k; QA PR #96; PPA PR #98

# Appendix D - piecewise formula

Y=alpha_context + gamma_degree + beta_0 ANY_VISIT + beta_plus EXTRA + error. ANY_VISIT=1(K>=1); EXTRA=max(min(K,7)-1,0). Z betas 0.30656/0.03609; PASS betas 0.11923/0.02225.

Source: two_part_attendance_z_pass.csv

# Appendix E - shape-selection sensitivity

| Grade-Z shape | In-sample AIC | Grouped held-out R2 skill | | Linear | 18753.43 | 0.01692 | | Logarithmic | 18753.99 | 0.01676 | | Square-root | 18753.62 | 0.01685 | | Saturating | 18754.40 | 0.01672 | | Categorical | 18762.27 | 0.01630 | The five fitted shapes differ little in held-out skill; professor-period fixed effects were unavailable for unseen validation contexts.

Source: attendance_curves/model_comparison_in_sample.csv and group_heldout_shape_cv.csv

# Appendix F - fused protocol

Discovery n=4,675 / 133 contexts, reserve n=1,952 / 57 contexts; 100/100 bootstrap retunings successful. Z selected 0|1 in 100%. PASS all-fused under one-SE-style heuristic but 128/128 by exhaustive discovery BIC.

Source: 2026-10-07_fused_lasso_100_bootstrap_results.md

# Appendix G - distributions

Numeric complete-case BIC prefers GMM K=2 vs single skew-normal in groups 0,1,2; skew-normal K=1 in 3,4,5,6+. Held-out log predictive density also varies. Do not infer underlying social subgroups.

Source: 47_complete_case_skewnormal_vs_gmm.csv; 50_complete_case_skewnormal_vs_gmm_cross_validation.csv

# Appendix H - provenance and readiness

Historical source SHAs: Paper 2.1 1546c594; fused 29810a86; mixture b1cace12; Shapley 5915976f. Scientific review drafts: methods/claims matrix PR #97, integrated manuscript PR #99; PPA PR #98 has code but no validated institutional fit. DO NOT RELEASE: QA PR #96 reports controlled student records in the public repository and invalid historical PASS-logit inference; privacy remediation and re-audit required. The deck PDF is an offline fallback preview, not an official Marp CLI-validated release.

Source: Source branch tips; PRs #96-#99; 2026-10-26